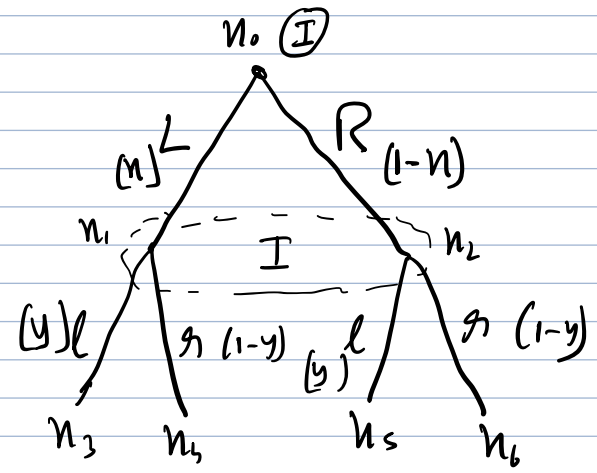
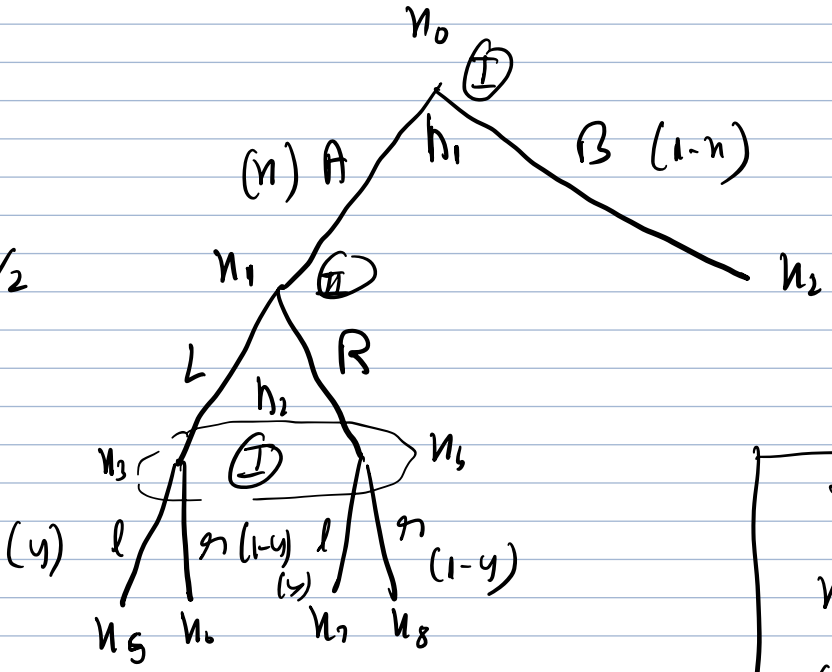
Class 25



$\sigma_1(Ll) = \sigma_1(Rr) = \frac{1}{2}$

$xy = \frac{1}{2}$, $x(1-y) = 0$, $(1-x)y = 0$, $(1-x)(1-y) = 1/2$



$\sigma_1(Bl) = 1/2$

$\sigma_1(Br) = 1/2$

$\sigma_1(Al) = 1/5$

$\sigma_1(Ar) = \frac{1}{10}$

$\sigma_1(Bl) = 2/5$

$\sigma_1(Br) = 3/10$

$xy = 1/5$

$x(1-y) = 1/10$

$(1-x)y = 2/5$

$(1-x)(1-y) = 3/10$

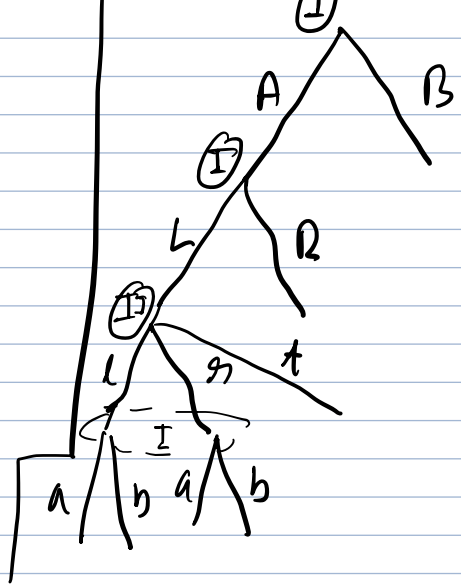
$S_1(h_2) \subseteq S_1$: The set of strategies of P-1 s.t. the game possibly reaches h_2.

Thm n, for every $s_1 \in S_1(h_2)$, $\exists s_2 \in S_2$ s.t. under the strategy profile (s_1, s_2) the game reaches h_2

$S_1(h_2) = \{Al, Ar\}$

$S_1(l; h_2) \subseteq S_1(h_2)$: the set of strategies in $S_1(h_2)$ that chooses action l at h_2.

$S_1(l; h_2) = \{Al\}$



$Al a$, $Al b$

$\sigma_1(S_1(h_2))$: the total probability of $S_1(h)$ under σ_1

$\sigma_1(S_1(l; h_2))$: the total probability of $S_1(l; h_2)$ under σ_1

$$b_1(l; h_2) = \frac{\sigma_1(S_1(l; h_2))}{\sigma_1(S_1(h_2))}$$ provided $\sigma_1(S_1(h_2)) > 0$

$\alpha \in [0, 1]$ o.w.

$$b_1(r; h_2) = \frac{\sigma_1(S_1(r; h_2))}{\sigma_1(S_1(h_2))}$$ provided $\sigma_1(S_1(h_2)) > 0$

$1 - \alpha$ o.w.

$S_1(h_1) = S_1$

$S_1(A; h_1) = \{Al, Ar\}$

$$b_1(A; h_1) = \frac{\sigma_1(S_1(A; h_1))}{\sigma_1(S_1)} = \frac{\sigma_1(Al) + \sigma_1(Ar)}{1}$$

$= 1/5 + 1/10$

Theorem (Kuhn (1957))

In every game in extensive form, if P-i has perfect recall, then for every mixed strategy of P-i, $\exists$ a behavior strategy that is equivalent to the mixed strategy.

Equilibria in behavior strategies

Theorem If all players in an extensive form game have perfect recall, then the game has an NE in behavior strategies.

Proof From the extensive form game, construct the corresponding strategic form game.
As the extensive form game is finite the corresponding strategic form game is finite too.

Now, the strategic form game has a mixed strategy NE, say $\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$. For each σ_i^*, construct b_i^*, an outcome equivalent strategy to σ_i^*, Then we claim $b^* = (b_1^*, \dots, b_n^*)$ is an NE.

$$U_i(\sigma^*) = U_i(b^*) \quad \forall i \in N$$

Assume for contradiction b^* is not an NE

$\exists k \in N$ $\quad U_k(b^*) < U_k(b_k, b_{-k}^*)$

$= U_k(\sigma_k, b_{-k}^*)$

$\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$

$\downarrow$

$(b_1^*, \sigma_2^*, \dots, \sigma_n^*)$

where σ_k is equivalent to b_k

$= U_k(\sigma_k, \sigma_{-k})$

$\leq U_k(\sigma^*)$, a contradiction.